Experimental and Computational Setup

Experimental and Computational Setup I

Numbered-table references point to the canonical manuscript; they are not standalone local deck numbers.

Core generation uses Python, NumPy, Pillow, and the standard library. The default seed is 0 and generation is offline. The generators are analytic in their typed parameters: the seed is recorded in manifests and study plans but does not yet alter artifact bytes; seed-driven stimulus randomization awaits a stochastic generator. PNG, GIF, WAV, and NPZ are available without ffmpeg; MP4 and muxed audiovisual output require both ffmpeg and ffprobe. Optional capabilities fail explicitly rather than silently changing the requested artifact.

Experimental and Computational Setup I

The deterministic validation suite executes real NumPy synthesis and real temporary-file media round trips. It checks finite normalized arrays, typed parameter boundaries, canonical hashes, dimensions, channel layouts, sample rates, frame counts, timebases, duration agreement, synchronization offsets, decoded stream facts, and registry/evidence consistency.

This is a software-validation protocol, not a psychophysical experiment. The tests establish repeatability under the declared software and dependency conditions; they do not establish replicability of an observer effect under a new display, playback chain, population, or task. That distinction is why the release reports both the exact environment contract and the missing observer evidence.

Experimental and Computational Setup I

The publication workflow writes 15 scientific PNG figures plus a separately provenance-recorded editorial cover, machine-readable source-data sidecars, a typed figure registry, 10 markdown tables, and a publication report. Each generated figure records its source function, seed, caption, alt text, claim level, and SHA-256 digest. Generated files are disposable; the tracked source is the generator and its tests.

The registry checks captions, source namespaces and file hashes; the strict release gate additionally regenerates every scientific panel and compares its data and raster bytes under recorded renderer, font and dependency identities. This is a current-code consistency oracle, not an independent scientific model or a substitute for inspecting the rendered pages. The cover validator checks real source bytes/dimensions and variant transforms. Historical prompt/model/ selection claims are

reported but not independently authenticated; caller-supplied images without a history are explicitly unrecorded.

The observer harness is a data-free design layer. It creates reproducibly randomized trial records, exports typed study plans, generates synthetic responses for contract tests, and exposes model templates. It does not collect or ship participant data. The publication atlas additionally runs `duckrabbit.synthetic.feature_observer`, an explicitly hand-specified feature model with no training corpus. Its canonical pairwise probabilities are serialized as model output and are useful for end-to-end orchestration tests; they are not a real vision-model estimate, a psychometric function, or an observer result. Human sensitivity is deferred until a calibrated, consented, preregistered study supplies response data and an analysis contract.